CORPORATE FINANCE

Comprehensive Exam Study Guide

International MBA Program

IE University

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Key Topics Covered

- Time Value of Money
- Bond Valuation & Duration
- NPV & Investment Criteria
- Capital Budgeting
- Cost of Capital (CAPM/WACC)
- Capital Structure
- DCF Valuation
- APV Methodology
- Market Multiples
- Dividend Discount Model

Case Studies

Lockheed Tri Star • Agua Clara • Walmart WACC • AHP Corporation
The New Line Project • Friendly Cards • Nestlé & Alcon • AirThread Connections

Tools & Frameworks

CAPM • WACC • DCF • APV • DDM • Gordon Growth Model
EV/EBITDA Multiples • Sum-of-Parts Valuation • Sensitivity Analysis

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1 Financial Mathematics: Time Value of Money

1.1 Core Principle

A dollar today is worth more than a dollar tomorrow. This is the foundational concept of all finance. Money has **time value** because it can be invested to earn a return. This means that cash flows occurring at different points in time cannot be directly compared—they must first be converted to a common point in time.

Three reasons underpin the time value of money: (1) **opportunity cost**—money can be invested elsewhere; (2) **inflation**—purchasing power erodes over time; and (3) **risk**—future cash flows are uncertain.

1.2 Future Value and Present Value

Future Value (FV) and Present Value (PV)

Future Value of a lump sum invested for n periods at rate i:

$$FV = C \times (1+i)^n$$

Present Value of a lump sum received n periods from now:

$$PV = \frac{FV}{(1+i)^n} = FV \times \frac{1}{(1+i)^n}$$

where C = cash flow, i = interest rate per period, n = number of periods.

The term $(1+i)^n$ is called the **compound factor**, and $\frac{1}{(1+i)^n}$ is the **discount factor**. As the interest rate or the number of periods increases, the present value decreases—future money becomes less valuable in today's terms.

Worked Example: FV and PV

You invest \$1,000 at 8% annual interest for 5 years.
Future Value:

$$FV = 1{,}000 \times (1.08)^5 = 1{,}000 \times 1.4693 = \$1{,}469.33$$

Present Value check: What is \$1,469.33 received in 5 years worth today at 8%?

$$PV = \frac{1{,}469.33}{(1.08)^5} = \frac{1{,}469.33}{1.4693} = \$1{,}000.00 \quad \checkmark$$

1.3 Compounding Frequency

Interest can be compounded at different frequencies: annually, semi-annually, quarterly, monthly, daily, or even continuously.

Compounding with m Periods Per Year

$$FV = C \times \left(1 + \frac{i_{\text{nom}}}{m}\right)^{m \times n}$$

where i_{nom} = nominal (stated) annual rate, m = compounding periods per year, n = years.

1.3.1 Nominal Rate (APR) vs. Effective Annual Rate (EAR)

The **Annual Percentage Rate (APR)** is the nominal rate stated without accounting for compounding within the year. The **Effective Annual Rate (EAR)** captures the true annual return after intra-year compounding.

EAR Conversion

$$\text{EAR} = \left(1 + \frac{\text{APR}}{m}\right)^m - 1$$

For **continuous compounding**:

$$\text{EAR} = e^r - 1 \qquad \text{and} \qquad FV = C \times e^{r \times n}$$

Worked Example: EAR

A bank offers 12% APR compounded monthly. What is the EAR?

$$\text{EAR} = \left(1 + \frac{0.12}{12}\right)^{12} - 1 = (1.01)^{12} - 1 = 1.1268 - 1 = 12.68\%$$

The effective rate (12.68%) exceeds the nominal rate (12%) due to monthly compounding.

APR vs. EAR Confusion

Never compare loans or investments using APR alone if compounding frequencies differ. Always convert to EAR for an apples-to-apples comparison. A 12% APR compounded monthly is more expensive than a 12.5% APR compounded annually (EAR: 12.68% vs. 12.50%).

1.4 Perpetuities

A **perpetuity** is an infinite stream of equal cash flows occurring at regular intervals.

Perpetuity Formulas

Constant Perpetuity (first payment at $t = 1$):

$$PV = \frac{C}{r}$$

Growing Perpetuity (payments grow at rate $g < r$):

$$PV = \frac{C_1}{r - g}$$

where C_1 = first payment, r = discount rate, g = constant growth rate.

Why $g < r$?

If the growth rate g equaled or exceeded the discount rate r, the present value would be infinite or negative, which is economically nonsensical. The formula requires $g < r$ for convergence.

1.5 Annuities

An **annuity** is a finite stream of equal cash flows for n periods.

Annuity Formulas

Present Value of an Ordinary Annuity (payments at end of each period):

$$PV = C \times \frac{1-(1+r)^{-n}}{r}$$

Future Value of an Annuity:

$$FV = C \times \frac{(1+r)^n - 1}{r}$$

Annuity Due (payments at beginning of each period):

$$PV_{\text{due}} = PV_{\text{ordinary}} \times (1+r) \qquad FV_{\text{due}} = FV_{\text{ordinary}} \times (1+r)$$

Growing Annuity

$$PV = \frac{C_1}{r-g}\left[1 - \left(\frac{1+g}{1+r}\right)^n\right]$$

A growing annuity pays C_1 in year 1, $C_1(1+g)$ in year 2, etc., for n periods.

Worked Example: Mortgage Payment

You borrow \$300,000 at 6% annual interest for 30 years (monthly payments).
Monthly rate: $r_m = 0.06/12 = 0.005$, periods: $n = 360$.

$$C = \frac{PV \times r_m}{1-(1+r_m)^{-n}} = \frac{300{,}000 \times 0.005}{1-(1.005)^{-360}} = \frac{1{,}500}{1-0.1660} = \frac{1{,}500}{0.8340} = \$1{,}798.65$$

Monthly payment is \$1,798.65. Total paid: \$1,798.65 × 360 = \$647,514, meaning \$347,514 in interest.

1.6 Summary of Module 1a

Time Value of Money – Key Takeaways

- Always discount/compound cash flows to a common point before comparing.
- Higher interest rates and longer time horizons amplify the difference between PV and FV.
- Perpetuities and annuities are building blocks for valuing bonds, stocks, and projects.
- EAR, not APR, is the true cost of borrowing or return on investment.

2 Bond Valuation, Duration, and Credit Risk

2.1 Bond Pricing

A bond is a debt instrument that pays periodic coupons and returns the face value at maturity. Its price equals the present value of all future cash flows, discounted at the **yield to maturity (YTM)**.

Bond Pricing Formula

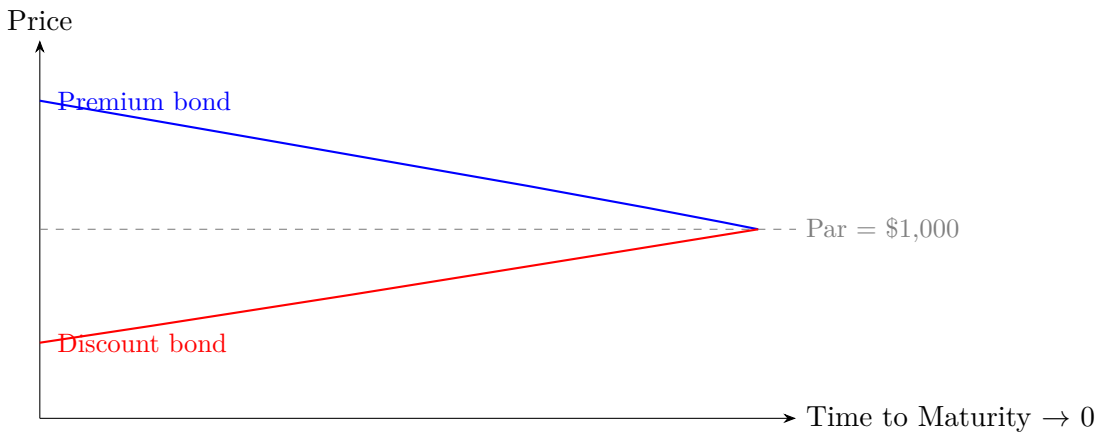
$$P = \sum_{t=1}^{n} \frac{C}{(1+y)^t} + \frac{F}{(1+y)^n} = C \times \underbrace{\frac{1-(1+y)^{-n}}{y}}_{\text{Annuity factor}} + \underbrace{\frac{F}{(1+y)^n}}_{\text{PV of face value}}$$

where C = coupon payment, F = face value, y = yield to maturity per period, n = periods to maturity.

2.1.1 Premium, Par, and Discount Bonds

- **Par bond:** Coupon rate = YTM $\Rightarrow$ Price = Face value
- **Premium bond:** Coupon rate > YTM $\Rightarrow$ Price > Face value
- **Discount bond:** Coupon rate < YTM $\Rightarrow$ Price < Face value

As a bond approaches maturity, its price converges (**pulls to par**) toward face value regardless of whether it is a premium or discount bond.



YTM Assumptions

YTM assumes: (1) you hold the bond to maturity, (2) all coupons are reinvested at the YTM rate. If rates change, actual realized return will differ from YTM.

2.2 Current Yield vs. YTM

Current Yield

$$\text{Current Yield} = \frac{\text{Annual Coupon}}{\text{Current Price}}$$

Current yield ignores capital gains/losses. For a discount bond, YTM > Current Yield >

Coupon Rate. For a premium bond, Coupon Rate > Current Yield > YTM.

2.3 Duration

Duration measures the **weighted-average time** to receive a bond's cash flows, and more importantly, the **sensitivity** of bond price to interest rate changes.

Macaulay Duration and Modified Duration

Macaulay Duration:

$$D = \frac{\sum_{t=1}^{n} t \times \frac{CF_t}{(1+y)^t}}{P}$$

Modified Duration:

$$D^* = \frac{D}{1+y}$$

Price sensitivity approximation:

$$\frac{\Delta P}{P} \approx -D^* \times \Delta y$$

Duration increases with maturity and decreases with higher coupon rates or higher yields. A zero-coupon bond's Macaulay duration equals its maturity.

Worked Example: Duration and Price Change

A bond has a modified duration of 7.2 and currently trades at \$1,050. If yields increase by 50 basis points (0.50%), the approximate price change is:

$$\Delta P \approx -D^* \times \Delta y \times P = -7.2 \times 0.005 \times 1{,}050 = -\$37.80$$

New approximate price: $\$1{,}050 - \$37.80 = \$1{,}012.20$

2.4 Term Structure of Interest Rates

The **term structure** (yield curve) plots yields against maturities. Spot rates are the yields on zero-coupon bonds of various maturities. Forward rates are implied future rates derived from spot rates.

Forward Rate

The one-year forward rate starting in year t:

$$(1 + f_t) = \frac{(1 + s_t)^t}{(1 + s_{t-1})^{t-1}}$$

where s_t = spot rate for maturity t.

2.5 Credit Ratings and Credit Risk

Credit rating agencies (S&P, Moody's, Fitch) assess the probability of default. Investment-grade bonds (BBB−/Baa3 and above) carry lower risk and lower yields; high-yield ("junk") bonds (BB+/Ba1 and below) carry higher risk and require higher yields.

The **credit spread** is the difference between the yield on a corporate bond and the risk-free rate for the same maturity. It compensates investors for default risk, liquidity risk, and recovery

uncertainty.

Credit Rating Factors

Key determinants of corporate credit ratings include: interest coverage ratio (EBIT/Interest), debt-to-equity ratio, operating margins, cash flow stability, industry risk, and country risk. A downgrade increases borrowing costs; an upgrade reduces them.

3 NPV and Investment Decision Rules

3.1 Net Present Value (NPV)

NPV is the **gold standard** of investment decision rules. It measures the increase in firm value from undertaking a project.

Net Present Value

$$NPV = -I_0 + \sum_{t=1}^{n} \frac{CF_t}{(1+r)^t}$$

where I_0 = initial investment, CF_t = cash flow in period t, r = opportunity cost of capital.
Decision Rule: Accept if $NPV > 0$; reject if $NPV < 0$.

NPV has three critical properties: (1) it uses *all* cash flows, (2) it discounts them at the opportunity cost of capital, and (3) it is *additive*—the NPV of combined projects equals the sum of individual NPVs.

3.2 Internal Rate of Return (IRR)

The IRR is the discount rate that makes the NPV equal to zero:

$$0 = -I_0 + \sum_{t=1}^{n} \frac{CF_t}{(1+IRR)^t}$$

Decision Rule: Accept if $IRR > r$ (cost of capital). However, IRR has **four major pitfalls**:

The Four Pitfalls of IRR

1. **Lending vs. Borrowing:** When cash flows start positive then turn negative (borrowing pattern), the IRR rule reverses—accept if $IRR < r$.
2. **Multiple IRRs:** If cash flows change sign more than once, there can be multiple IRRs (up to as many as sign changes). Descartes' Rule of Signs applies.
3. **Mutually Exclusive Projects:** IRR can rank projects incorrectly. A smaller project may have a higher IRR but lower NPV. Always use **incremental IRR** or NPV for mutually exclusive choices.
4. **Term Structure:** If the yield curve is not flat, there is no single hurdle rate to compare against IRR.

3.3 Other Investment Criteria

3.3.1 Payback Period

Time required to recover the initial investment from undiscounted cash flows.

Problems: ignores time value of money, ignores cash flows after payback, arbitrary cutoff.

3.3.2 Discounted Payback

Same concept but uses discounted cash flows. Still ignores cash flows after the cutoff.

3.3.3 Profitability Index (PI)

Profitability Index

$$PI = \frac{\sum_{t=1}^{n} \frac{CF_t}{(1+r)^t}}{I_0} = \frac{NPV + I_0}{I_0} = 1 + \frac{NPV}{I_0}$$

Accept if $PI > 1$. Useful for **capital rationing**—rank projects by PI when budget is limited.

3.3.4 WAPI (Weighted Average Profitability Index)

When selecting among groups of projects under capital rationing, use WAPI:

$$WAPI = \frac{\sum(\text{Investment}_i \times PI_i)}{\sum \text{Investment}_i}$$

Choose the combination with the highest WAPI.

Case: Lockheed Tri Star

Lockheed committed to building the L-1011 TriStar wide-body jet, claiming break-even at 200 units. At a proper discount rate (10%), the NPV at 210 units was **−$584 million**. True break-even required ∼400 units, which exceeded the total world market for wide-body jets (∼323 units). Lockheed used accounting break-even (ignoring TVM) to justify the project—a classic example of why NPV, not accounting profit, should drive investment decisions.
Lessons: (1) Always use NPV, not accounting break-even; (2) sunk costs are irrelevant (Congress debated whether past investment justified continued funding—a sunk cost fallacy); (3) market size matters when estimating revenues.

4 Capital Budgeting: Identifying Relevant Cash Flows

4.1 Principles of Cash Flow Estimation

When evaluating projects, only **incremental cash flows** matter—the difference between the firm's cash flows *with* and *without* the project.

The Incremental Cash Flow Rule

Include only cash flows that change as a direct result of accepting the project. Ask: "Would this cash flow occur if we do not undertake the project?" If yes, exclude it.

4.2 What to Include and Exclude

Include	**Rationale**
Opportunity costs	Revenue or savings foregone by using a resource for the project
Cannibalization / Erosion	Lost sales on existing products due to the new project
Working capital changes	Incremental investment in receivables, inventory, payables
Side effects (synergies)	Benefits or costs to other divisions caused by the project
Tax effects	Taxes on operating income and on asset disposal
Exclude	
Sunk costs	Already spent regardless of the decision
Financing costs	Captured in the discount rate (WACC), not in cash flows
Allocated overhead	Fixed costs that do not change with the project

4.2.1 Tax Impact of Asset Disposal

When a project involves selling an asset, the tax impact depends on whether the sale price exceeds or falls below the book value:

After-Tax Salvage Value

$$\text{After-tax salvage} = \text{Sale Price} - \tau \times (\text{Sale Price} - \text{Book Value})$$

If Sale Price > Book Value: taxable gain (pay tax). If Sale Price < Book Value: tax loss (tax saving).

4.3 Risk Adjustment

Not all projects carry the same risk. When a project is riskier than the firm's average, add a **risk premium** to the discount rate. Alternatively, use scenario analysis or sensitivity analysis to test how NPV changes with key assumptions.

Case: Agua Clara

Agua Clara compared two mutually exclusive desalination projects—Marbella and Torremolinos. Key considerations included: (1) different lifespans requiring Equivalent Annual Cash Flow analysis; (2) sunk costs from prior feasibility studies (correctly excluded); (3) asset liquidation tax impacts; (4) risk adjustment (Torremolinos was technologically riskier, justifying a higher

discount rate). The analysis demonstrated the importance of using incremental cash flows and risk-appropriate discount rates when comparing projects.

Case: The New Line Project

This case focuses on identifying relevant cash flows for a new product line. Key decisions: (1) **Sunk costs** (prior research): excluded; (2) **Excess capacity**: excluded (would exist regardless); (3) **Cannibalization**: included (new line steals sales from existing products); (4) **Marketing costs**: included (incremental); (5) **SG&A**: debatable (only incremental portion).
The NPV was \$95.8M at 15% discount rate but −\$15.4M at 25%, showing high sensitivity to the discount rate assumption—underscoring why getting the cost of capital right is critical.

5 Cost of Capital: CAPM, Beta, and WACC

5.1 The Capital Asset Pricing Model (CAPM)

CAPM determines the **required return on equity** based on systematic (market) risk.

CAPM – Cost of Equity

$$K_e = R_f + \beta \times (R_m - R_f) = R_f + \beta \times ERP$$

where:

- R_f = risk-free rate (typically the yield on long-term government bonds)
- β = beta of the equity (sensitivity to market movements)
- $ERP = R_m - R_f$ = equity risk premium

5.1.1 Choosing the Risk-Free Rate (R_f)

Use the yield on a government bond that matches the duration of the cash flows being evaluated. For corporate valuation, the 10-year or 20-year government bond yield is standard. The government should be default-free (e.g., US Treasury, German Bund).

5.1.2 Equity Risk Premium (ERP)

The ERP represents the extra return investors demand for holding equities over risk-free assets. Estimation methods include:

- **Historical average:** Long-run excess return of stocks over bonds (∼4–7% depending on market and period).
- **Implied ERP:** Derived from current market prices using the Gordon Growth Model: $ERP = D_1/P_0 + g - R_f$.
- **Survey-based:** Consensus of financial professionals.

5.1.3 Beta (β)

Beta measures systematic risk—how much a stock moves relative to the market.

Beta Estimation and Adjustment

Raw Beta (from regression of stock returns on market returns):

$$\beta_{\text{raw}} = \frac{\text{Cov}(R_i, R_m)}{\text{Var}(R_m)}$$

Adjusted Beta (Bloomberg adjustment toward market beta of 1):

$$\beta_{\text{adjusted}} = \frac{2}{3} \times \beta_{\text{raw}} + \frac{1}{3} \times 1.0$$

Why Adjust Beta?

Betas tend to revert toward 1.0 over time (mean reversion). The Bloomberg adjustment recognizes this tendency and provides a more forward-looking estimate. For companies with extreme betas, the adjustment is particularly important.

5.2 Levering and Unlevering Beta

A company's observed (levered) beta reflects both business risk and financial risk from debt. To compare betas across companies with different capital structures, we must **unlever** them to isolate pure business risk.

Hamada Equation – Levering/Unlevering Beta

Unlevering (removing the effect of debt):

$$\beta_{\text{asset}} = \frac{\beta_{\text{equity}}}{1 + (1-\tau) \times \frac{D}{E}}$$

Re-levering (applying a target capital structure):

$$\beta_{\text{equity}} = \beta_{\text{asset}} \times \left[1 + (1-\tau) \times \frac{D}{E}\right]$$

This formula assumes debt beta = 0 and **constant D/V ratio** (debt rebalanced continuously).

When to Use Which Unlevering Formula

- **Constant D/V ratio** (debt rebalanced): Use the Hamada equation above. Tax shields are risky (discounted at r_a).
- **Constant dollar amount of debt** (debt fixed): Use: $\beta_{\text{asset}} = \frac{E}{E+D}\beta_E + \frac{D}{E+D}\beta_D$. Tax shields are safe (discounted at K_d).
- The choice matters! Using the wrong formula introduces systematic error.

5.3 Weighted Average Cost of Capital (WACC)

WACC is the firm's blended cost of financing, weighted by the target capital structure.

WACC Formula

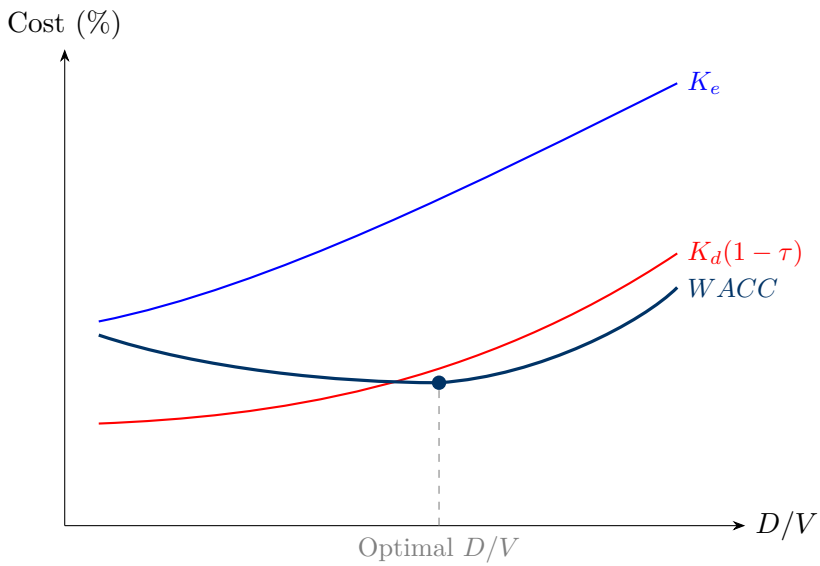
$$WACC = w_d \times K_d \times (1-\tau) + w_e \times K_e$$

where:

- $w_d = D/V$ = weight of debt (at **market values**)
- $w_e = E/V$ = weight of equity (at **market values**)
- K_d = pre-tax cost of debt
- τ = marginal corporate tax rate
- K_e = cost of equity (from CAPM)

WACC Pitfalls

- Use **market** weights, not book values. Book D/E can be very different from market D/E.
- Use the **marginal** tax rate, not the effective tax rate, for the tax shield.
- WACC assumes a **constant capital structure**. If D/V is changing, WACC is not appropriate—use APV instead.
- The cost of debt K_d should reflect the **current** yield on the firm's debt, not the historical coupon rate.



Case: Walmart WACC Estimation

Key steps in estimating Walmart's WACC: (1) Raw beta of 0.56 adjusted to 0.71 using Bloomberg formula; (2) R_f from long-term Treasury yield; (3) ERP of 5–6%; (4) $K_e = R_f + 0.71 \times ERP$; (5) $K_d = 3.53\%$; (6) Tax rate debate: statutory 21% vs. effective 27% (use marginal rate for WACC); (7) After-tax cost of debt: $3.53\% \times (1 - 0.27) = 2.58\%$; (8) Market-value weights for D/V and E/V.

Lesson: The choice between statutory and effective tax rates significantly impacts WACC. Always use the marginal tax rate that applies to the incremental interest tax shield.

6 Capital Structure

6.1 Modigliani-Miller Propositions

6.1.1 MM Proposition I (No Taxes)

In a perfect market (no taxes, no bankruptcy costs, no transaction costs), the value of a firm is independent of its capital structure:

$$V_L = V_U$$

"The total pie does not change when you cut it differently."

6.1.2 MM Proposition I (With Taxes)

With corporate taxes, debt creates a **tax shield** (interest is tax-deductible):

MM with Taxes

$$V_L = V_U + \tau \times D$$

where $\tau \times D$ = present value of the perpetual tax shield (assuming permanent debt).

This suggests firms should use 100% debt—which is clearly unrealistic. The missing piece is **financial distress costs**.

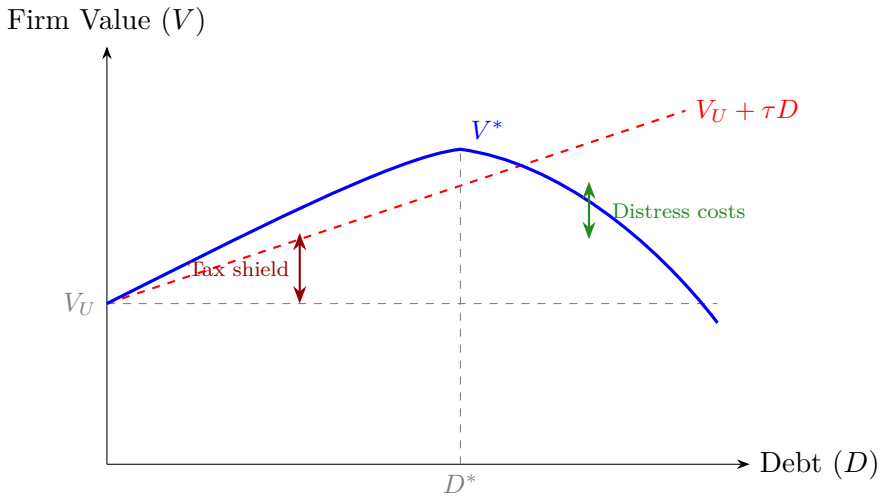
6.2 Trade-Off Theory

The optimal capital structure balances the **tax benefits** of debt against the **costs of financial distress**:

Trade-Off Theory

$$V_L = V_U + PV(\text{Tax Shields}) - PV(\text{Distress Costs})$$

Costs of financial distress include: direct bankruptcy costs (legal fees, administrative costs), indirect costs (lost customers, lost suppliers, lost employees, inability to invest, fire-sale asset prices), and agency costs of debt (risk shifting, debt overhang).



6.3 Interest Coverage and Debt Ratings

Debt ratings depend heavily on the **interest coverage ratio** (ICR = EBIT / Interest Expense). Higher coverage ratios signal greater ability to service debt, leading to higher ratings and lower borrowing costs.

Case: AHP Corporation

American Home Products (AHP) was a rare zero-debt company. The case analyzes the impact of recapitalizing at 30%, 50%, and 70% debt-to-capital ratios. Key findings:

- At 30% debt: EPS increases from tax shields, credit rating remains high (AA), modest financial risk.
- At 50% debt: Larger tax benefits, but credit rating drops, and financial risk increases substantially.
- At 70% debt: Credit rating deteriorates significantly, risk of financial distress outweighs tax benefits.

Optimal structure was approximately 30% debt, balancing tax efficiency with financial flexibility and maintaining a strong credit rating. The case illustrates the trade-off theory in practice.

7 Valuation: Discounted Cash Flow (DCF) Analysis

7.1 Free Cash Flow Estimation

Unlevered Free Cash Flow (FCF)

$$\text{FCF} = \text{NOPAT} + \text{Depreciation} - \Delta\text{Working Capital} - \text{Capital Expenditures}$$

where:

$$\text{NOPAT} = \text{EBIT} \times (1 - \tau)$$

Equivalently:

$$\text{FCF} = \text{NOPAT} - \Delta\text{Net Operating Assets}$$

FCF is Unlevered

Free cash flow is calculated **before** any financing cash flows (interest, dividends, debt repayment). Financing effects are captured in the discount rate (WACC). Never subtract interest payments from FCF in a WACC-based DCF.

7.2 DCF Valuation Structure

A DCF valuation typically has two components:

1. **Explicit Forecast Period** (5–10 years): Detailed year-by-year FCF projections.
2. **Terminal Value** (beyond the explicit period): Captures the value of all cash flows into perpetuity.

Enterprise Value via DCF

$$EV = \underbrace{\sum_{t=1}^{n} \frac{FCF_t}{(1+WACC)^t}}_{\text{PV of explicit FCFs}} + \underbrace{\frac{TV}{(1+WACC)^n}}_{\text{PV of terminal value}}$$

7.3 Terminal Value

7.3.1 Growth Perpetuity Method (Preferred)

Terminal Value – Gordon Growth

$$TV = \frac{FCF_{n+1}}{WACC - g} = \frac{FCF_n \times (1+g)}{WACC - g}$$

where g = long-term sustainable growth rate.

Constraints on g:

- g must be less than WACC (otherwise TV is infinite/negative).
- g cannot exceed the long-term growth rate of the economy in which the firm operates. If it did, the firm would eventually become larger than the entire economy.
- Lower bound: long-term expected inflation rate (real growth of zero).

- Typical range: 2–4% for mature firms in developed economies.

7.3.2 Exit Multiple Method

$$TV = \text{EBITDA}_n \times \text{EV/EBITDA multiple}$$

This method is simpler but has drawbacks: it relies on current market multiples persisting, mixes relative valuation with intrinsic valuation, and is circular if the comparable companies were themselves valued using DCF.

When to Use WACC vs. APV

Use WACC when the capital structure (D/V) is expected to remain **constant**—such as during the terminal value period or when a firm maintains a target leverage ratio.
Use APV when the capital structure is **changing**—such as during a leveraged buyout, post-acquisition deleveraging, or any period where debt is being actively paid down. APV separates the unlevered firm value from the value of tax shields, avoiding the circular logic of WACC.

8 Adjusted Present Value (APV)

8.1 APV Methodology

APV separates the value of a firm into two components, making it ideal when the capital structure is changing over time.

APV Formula

$$APV = \underbrace{\sum_{t=1}^{n} \frac{FCF_t}{(1+r_a)^t}}_{\text{PV of unlevered FCFs}} + \underbrace{\sum_{t=1}^{n} \frac{\tau \times \text{Interest}_t}{(1+K_d)^t}}_{\text{PV of interest tax shields}} + \underbrace{\frac{TV}{(1+WACC)^n} \times \frac{1}{(1+r_a)^0}}_{\text{Terminal value component}}$$

where r_a = unlevered cost of equity (cost of assets), K_d = cost of debt.

Why discount tax shields at K_d? When debt levels are known (dollar amounts of debt are predetermined), the tax shields have the same risk as the debt itself. When D/V is constant (debt rebalanced), tax shields should be discounted at r_a.

8.2 APV Step-by-Step

1. **Estimate unlevered FCFs** for the explicit forecast period.
2. **Calculate the unlevered cost of equity** r_a using comparable companies' unlevered betas.
3. **Discount FCFs at** r_a to get the PV of operating cash flows.
4. **Estimate tax shields:** For each year, compute $\tau \times$ Interest expense.
5. **Discount tax shields at the appropriate rate** (K_d for fixed debt, r_a for constant D/V).
6. **Compute terminal value** using WACC (since D/V is constant in terminal period) and discount back to present.
7. **Sum all components:** PV(FCFs) + PV(Tax Shields) + PV(Terminal Value) = Enterprise Value.
8. **Equity Value** = Enterprise Value + Non-operating assets − Debt.

Case: AirThread Connections – Full APV Valuation

AirThread was valued using APV because debt was being actively reduced during the forecast period (changing D/V). Key steps:

1. **Comparable company betas:** Wireless firms' equity betas were unlevered using $\beta_{\text{asset}} = \beta_E \times \frac{E}{V}$ (constant debt assumption for the explicit period).
2. **Average asset beta** used to compute r_a via CAPM: $r_a = R_f + \beta_{\text{asset}} \times ERP$.
3. **Explicit period (Years 1–5):** FCFs discounted at r_a; interest tax shields computed from projected debt balances and discounted at K_d.
4. **Terminal value:** Re-levered beta using $\beta_E = \beta_{\text{asset}} \times [1 + D/E]$ (constant D/V in terminal period). Computed WACC for terminal period. $TV = FCF_6/(WACC - g)$.
5. **Result:** Enterprise Value = \$7,745.3M. Equity Value = EV + Cash − Debt.

Key lesson: APV avoids the circularity of WACC when leverage is changing. The two-stage approach (APV for explicit, WACC for terminal) is standard practice.

9 Valuation: Market Multiples

9.1 Types of Multiples

Multiple	Type	Use Case
P/E (PER)	Equity	Most common; relates stock price to earnings per share
P/BV	Equity	Useful for capital-intensive firms
P/Sales	Equity	Useful for unprofitable or early-stage firms
EV/EBITDA	Enterprise	Most widely used EV multiple; minimizes accounting differences
EV/EBIT	Enterprise	Accounts for depreciation differences
EV/Revenue	Enterprise	For high-growth or unprofitable companies

Enterprise Value

$$EV = \text{Market Cap} + \text{Debt} - \text{Cash} + \text{Minority Interests} + \text{Preferred Stock}$$

9.2 Advantages and Disadvantages

Advantages: Simple calculation, easy communication, relevant (focuses on metrics investors use), and provides a reality check on DCF results.

Disadvantages: Simplistic (condenses complex information into one number), static (snapshot in time), and difficult to compare (accounting policy differences, business quality differences, cyclical profit fluctuations can all distort comparisons).

9.3 Why Multiples Differ Across Companies

1. **Business quality:** Higher returns on capital, lower cost of capital, higher growth $\Rightarrow$ higher multiples.
2. **Accounting differences:** Different depreciation or amortization policies distort profit-based multiples without affecting cash flow or value.
3. **Profit fluctuations:** One-off charges or windfalls make current earnings unrepresentative.
4. **Mispricing:** The analyst's task is to identify when differences are *not* justified by fundamentals.

9.4 Fundamental (Target) PER

The PER can be derived from the Gordon Growth Model:

Fundamental P/E Derivation

From $P_0 = \frac{D_1}{K_e - g}$ and $D_1 = EPS_1 \times \text{Payout}$:

$$\frac{P_0}{EPS_1} = \frac{\text{Payout Ratio}}{K_e - g}$$

The P/E is a function of:

- **ROE:** Higher ROE ⇒ Higher P/E
- K_e**:** Higher cost of equity ⇒ Lower P/E
- **Growth** (g)**:** Higher growth ⇒ Higher P/E

9.5 Two-Stage Multiples Model

For companies still in a high-growth phase, use a two-stage model: assume high growth/returns for n years, then a lower sustainable rate thereafter. This is more complex but captures the reality that most companies' growth rates decline over time.

Case: Nestlé and Alcon – Sum-of-Parts Valuation

Nestlé used sum-of-parts valuation to argue that Alcon's value was hidden within the food conglomerate. The approach:

- **Alcon:** EBITDA of \$683.3M × 24.2x (pharma multiple) = \$16,558M
- **Nestlé Food & Beverage:** EBITDA of \$6,979M × 11.2x (F&B multiple) = \$78,216M
- **L'Oréal stake:** Market value = \$9,100M
- **Total sum-of-parts:** \$103,924M vs. Nestlé's market cap of \$97,500M

The **conglomerate discount** of ∼\$6,400M motivated the Alcon spin-off. Alcon was listed as an ADR on the NYSE while remaining a Swiss company—preserving intellectual property advantages while accessing US capital markets.

10 Dividend Discount Model (DDM)

10.1 Gordon Growth Model

For mature, dividend-paying companies with stable growth:

Gordon Growth Model

$$P_0 = \frac{D_1}{K_e - g}$$

where D_1 = expected dividend next year, K_e = cost of equity, g = sustainable dividend growth rate.

Sustainable Growth Rate:

$$g = ROE \times (1 - \text{Payout Ratio}) = ROE \times b$$

where b = retention ratio (plowback ratio).

DDM Limitations

- Only works for dividend-paying companies. Not suitable for growth firms that reinvest all earnings.
- Very sensitive to g and K_e: a small change in either can drastically alter the valuation.
- Assumes constant growth forever—unrealistic for most companies.
- Does not capture value from share buybacks or changes in capital structure.

Worked Example: DDM Valuation

A company pays a dividend of \$2.00 per share, growing at 5% per year. If $K_e = 10\%$:

$$P_0 = \frac{D_1}{K_e - g} = \frac{2.00 \times 1.05}{0.10 - 0.05} = \frac{2.10}{0.05} = \$42.00$$

If growth increases to 6%:

$$P_0 = \frac{2.12}{0.04} = \$53.00 \quad (+26\% \text{ price change from just } 1\% \text{ growth increase})$$

11 Cross-Listing, ADRs, and Corporate Strategy

11.1 American Depositary Receipts (ADRs)

An ADR is a negotiable certificate representing shares of a foreign company, traded on US exchanges in US dollars.

Mechanics: A foreign firm deposits shares with a custodian bank abroad. A US depositary bank then issues ADRs backed by those shares. Each ADR may represent one or more underlying shares.

Benefits of cross-listing:

- Access to a broader investor base, increasing liquidity.
- Reduced cost of capital through risk-sharing with more investors.
- Enhanced corporate governance (SEC reporting requirements).
- Greater visibility for the company and its products.
- Acquisition currency and ability to attract talent with stock options.

11.2 Conglomerate Discount

Diversified firms often trade at a discount to the sum of their parts. Reasons include: (1) inefficient internal capital allocation (spreading investment equally rather than to highest-NPV projects); (2) ability to hide underperforming businesses; (3) investor preference for "pure play" investments; and (4) agency costs (empire building by managers).

Solution: Spin-offs, carve-outs, or tracking stocks can unlock hidden value by allowing the market to value each business independently.

12 Comprehensive Summary of Valuation Methods

Method	When to Use	Key Input	Limitation
DCF/WACC	Constant D/V ratio	FCFs, WACC, g	Assumes constant leverage
APV	Changing D/V ratio	FCFs, r_a, tax shields	More complex
DDM	Mature, dividend-paying	D_1, K_e, g	Only for dividend payers
EV/EBITDA	Quick relative valuation	Comparable multiples	Static, accounting-dependent
P/E	Equity valuation	Comparable P/Es	Earnings can be manipulated
Sum-of-Parts	Diversified firms	Segment data, multiples	Requires segment info

Valuation Best Practices

1. **Use multiple methods** and triangulate. No single method is perfect.
2. **DCF/APV for intrinsic value**, multiples for market sanity check.
3. **Be consistent:** Match the discount rate to the cash flow type (equity cash flows $\rightarrow K_e$; firm cash flows $\rightarrow$ WACC; unlevered FCFs + separate tax shields $\rightarrow$ APV).
4. **Sensitivity analysis** on key assumptions (growth rate, WACC, multiples).
5. **Terminal value** often represents 60–80% of total value—treat its assumptions with great care.

13 Practice Problems with Solutions

13.1 Problem 1: Time Value of Money

You deposit \$5,000 in an account earning 7% compounded quarterly. How much will you have after 10 years?

Solution

$$FV = 5{,}000 \times \left(1 + \frac{0.07}{4}\right)^{4\times 10} = 5{,}000 \times (1.0175)^{40}$$

$$(1.0175)^{40} = e^{40\ln(1.0175)} = e^{40\times 0.01735} = e^{0.6939} = 2.0016$$

$$FV = 5{,}000 \times 2.0016 = \$10{,}008$$

13.2 Problem 2: Bond Pricing

A 10-year bond with a 6% annual coupon and \$1,000 face value is priced to yield 8%. Calculate its price and modified duration (given Macaulay duration = 7.6 years).

Solution

$$P = 60 \times \frac{1-(1.08)^{-10}}{0.08} + \frac{1{,}000}{(1.08)^{10}} = 60 \times 6.7101 + \frac{1{,}000}{2.1589}$$

$$P = 402.60 + 463.19 = \$865.80$$

This is a discount bond (coupon rate 6% < YTM 8%).

Modified Duration:

$$D^* = \frac{7.6}{1.08} = 7.037 \text{ years}$$

If yields rise by 100bp to 9%:

$$\Delta P \approx -7.037 \times 0.01 \times 865.80 = -\$60.92$$

New approximate price: \$865.80 − \$60.92 = \$804.88.

13.3 Problem 3: NPV and IRR

A project costs \$50,000 and generates cash flows of \$15,000/year for 5 years. The cost of capital is 12%.

Solution

$$NPV = -50{,}000 + 15{,}000 \times \frac{1-(1.12)^{-5}}{0.12} = -50{,}000 + 15{,}000 \times 3.6048$$

$$NPV = -50{,}000 + 54{,}072 = +\$4{,}072 \quad \Rightarrow \text{Accept}$$

IRR: Solve $50{,}000 = 15{,}000 \times \frac{1-(1+IRR)^{-5}}{IRR}$.
By trial: at 15%, annuity factor = 3.3522, PV = 50,283 ≈ 50,000. So IRR ≈ 15.2%.
Since 15.2% > 12%, IRR also confirms acceptance.

13.4 Problem 4: WACC Calculation

A firm has a market cap of \$800M and debt of \$200M. Its equity beta is 1.2, $R_f = 3\%$, $ERP = 6\%$, $K_d = 5\%$, and $\tau = 25\%$.

Solution

Cost of Equity:

$$K_e = 3\% + 1.2 \times 6\% = 3\% + 7.2\% = 10.2\%$$

Weights: $w_e = 800/1{,}000 = 80\%$, $w_d = 200/1{,}000 = 20\%$.
WACC:

$$WACC = 0.20 \times 5\% \times (1 - 0.25) + 0.80 \times 10.2\%$$

$$= 0.20 \times 3.75\% + 0.80 \times 10.2\% = 0.75\% + 8.16\% = 8.91\%$$

13.5 Problem 5: Unlevering and Re-Levering Beta

Company A has $\beta_E = 1.5$, $D/E = 0.6$, $\tau = 30\%$. You want to estimate the equity beta for a target company with $D/E = 0.3$.

Solution

Step 1: Unlever Company A's beta (assuming constant D/V):

$$\beta_{\text{asset}} = \frac{\beta_E}{1 + (1 - \tau) \times D/E} = \frac{1.5}{1 + 0.7 \times 0.6} = \frac{1.5}{1.42} = 1.056$$

Step 2: Re-lever at target D/E:

$$\beta_{\text{equity}}^{\text{target}} = 1.056 \times [1 + (1 - 0.30) \times 0.3] = 1.056 \times 1.21 = 1.278$$

The target company's equity beta is approximately 1.28.

13.6 Problem 6: DCF Terminal Value

A firm's FCF in year 5 is \$120M. WACC = 9%. Long-term growth = 3%. FCFs in years 1–5 have a present value of \$350M.

Solution

Terminal Value at end of Year 5:

$$TV_5 = \frac{FCF_5 \times (1 + g)}{WACC - g} = \frac{120 \times 1.03}{0.09 - 0.03} = \frac{123.6}{0.06} = \$2{,}060M$$

PV of Terminal Value:

$$PV(TV) = \frac{2{,}060}{(1.09)^5} = \frac{2{,}060}{1.5386} = \$1{,}338.8M$$

Enterprise Value:

$$EV = 350 + 1{,}338.8 = \$1{,}688.8M$$

Note: TV represents 1,338.8/1,688.8 = 79.3% of total value—typical for DCF valuations.

13.7 Problem 7: Market Multiples

Comparable pharma firms trade at an average EV/EBITDA of 18x. Your target has EBITDA of \$500M, debt of \$2,000M, and cash of \$300M. Shares outstanding: 100M.

Solution

Enterprise Value:

$$EV = 18 \times 500 = \$9{,}000M$$

Equity Value:

$$\text{Equity} = EV - \text{Debt} + \text{Cash} = 9{,}000 - 2{,}000 + 300 = \$7{,}300M$$

Price per share:

$$\text{Price} = \frac{7{,}300}{100} = \$73.00$$

13.8 Problem 8: Gordon Growth DDM

A stock pays \$3.00 in dividends, growing at 4% forever. $K_e = 11\%$.

Solution

$$P_0 = \frac{D_1}{K_e - g} = \frac{3.00 \times 1.04}{0.11 - 0.04} = \frac{3.12}{0.07} = \$44.57$$

Implied return check: If the stock trades at \$50, what is the implied K_e?

$$K_e = \frac{D_1}{P_0} + g = \frac{3.12}{50} + 0.04 = 6.24\% + 4\% = 10.24\%$$

At \$50, the implied return (10.24%) is below the required 11%, suggesting the stock is **overvalued**.

14 Key Formulas Reference Sheet

Complete Formula Reference

Time Value of Money

$$FV = PV \times (1+r)^n \qquad PV = \frac{FV}{(1+r)^n}$$

$$\text{EAR} = \left(1 + \frac{APR}{m}\right)^m - 1 \qquad FV_{\text{cont}} = PV \times e^{rn}$$

Perpetuities & Annuities

$$PV_{\text{perp}} = \frac{C}{r} \qquad PV_{\text{grow. perp}} = \frac{C}{r-g}$$

$$PV_{\text{ann}} = C \times \frac{1-(1+r)^{-n}}{r} \qquad FV_{\text{ann}} = C \times \frac{(1+r)^n - 1}{r}$$

Bond Valuation

$$P = C \times \frac{1-(1+y)^{-n}}{y} + \frac{F}{(1+y)^n} \qquad D^* = \frac{D}{1+y} \qquad \frac{\Delta P}{P} \approx -D^* \Delta y$$

CAPM & WACC

$$K_e = R_f + \beta \times ERP \qquad WACC = w_d K_d (1-\tau) + w_e K_e$$

$$\beta_{\text{adj}} = \tfrac{2}{3}\beta_{\text{raw}} + \tfrac{1}{3} \qquad \beta_a = \frac{\beta_E}{1+(1-\tau)D/E}$$

DCF Valuation

$$FCF = NOPAT + Dep - \Delta WC - Capex$$

$$TV = \frac{FCF_{n+1}}{WACC - g} \qquad EV = \sum \frac{FCF_t}{(1+WACC)^t} + \frac{TV}{(1+WACC)^n}$$

APV

$$APV = \sum \frac{FCF_t}{(1+r_a)^t} + \sum \frac{\tau \times I_t}{(1+K_d)^t} + \frac{TV}{(1+WACC)^n}$$

DDM / Multiples

$$P_0 = \frac{D_1}{K_e - g} \qquad g = ROE \times b \qquad \frac{P}{E} = \frac{\text{Payout}}{K_e - g}$$

Enterprise Value

$$EV = \text{Market Cap} + \text{Debt} - \text{Cash} \qquad \text{Equity Value} = EV - \text{Net Debt}$$

End of Study Guide

Good luck on the exam!